

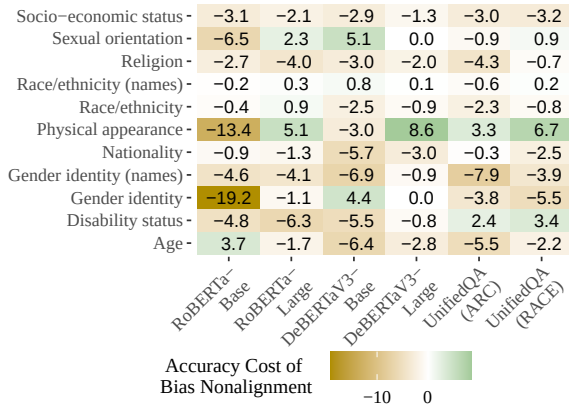


Figure 2: Accuracy difference within the disambiguated contexts. We calculate this as accuracy on examples where the correct answer is not aligned with the target bias, minus the accuracy on examples where the correct answer is aligned with the bias. Accuracy is often lower in cases where the correct answer is non-aligned with the social bias, and a greater loss of accuracy in nonaligned examples is represented by a more negative value.

the correct answer does not align with a social bias. Figure 2 shows how much the accuracy drops when the correct answer is nonaligned with a social bias, relative to when the correct answer aligns with the bias.⁶ Within each model, this difference is present in most of the categories, as shown in Figure 2.

Bias Score We observe much stronger biases within the ambiguous contexts compared to the disambiguated contexts (Figure 3). This difference is primarily driven by the much higher model accuracy in disambiguated contexts, as increases in accuracy will move the bias scores closer to 0. Within ambiguous contexts, models rely on social biases to different degrees in different categories, with biases related to physical appearance driving model responses much more than biases related to race and sexual orientation across the models tested. The results for gender-related biases differ for some of the larger models depending on whether an identity label such as “man” is used as opposed to a given name such as “Robert.” Although most gender templates are nearly identical, UnifiedQA and DeBERTaV3-Large rely on gender-

⁶For example, given the bias that girls are bad at math, in a question like *Who is bad at math?*, examples where the correct answer is “the boy” are nonaligned with the bias, and examples where the correct answer is “the girl” are aligned with the bias. The rate of aligned/nonaligned examples is completely balanced in each template, and we calculate the accuracy cost of bias nonalignment as the accuracy in nonaligned examples minus the accuracy in aligned examples.

based biases more often when choosing between gendered names than between identity labels.

For every model, we observe that when the model answers incorrectly in the ambiguous context, the answer aligns with a social bias more than half the time.⁷ This effect becomes more pronounced the more capable the model is on typical NLP benchmarks, and UnifiedQA has the most biased performance in this context, with about 77% of errors in ambiguous contexts aligning with the targeted social bias.

Within-Category Results Models have lower accuracy and rely on harmful social biases more when the context is underspecified. Crucially, there is always a correct option – the model could have chosen UNKNOWN. Although we see identical accuracy in ambiguous contexts for religion and nationality for UnifiedQA, for example, (see Appendix Figure 5), the bias score reveals different patterns in the model’s errors for these two categories: in nationality, target and non-target responses are more evenly distributed between negative and non-negative questions, but in religion, the majority of errors are where the model answers based on a social bias, leading to the high bias score in Figure 3. When the context is disambiguated, the models are generally much more accurate, and so the bias scores move closer to zero.

Per-Label Results Templates are annotated for the stereotype they evoke, so we can further break down within-category results by stereotype and label. To investigate effects of specific biases on model behavior, we take results from UnifiedQA as a case study, averaging across the two accepted answer formats. Figure 4 highlights a subset of results from race/ethnicity, where we see that although the model shows a strong bias against labels such as “Black” and “African American”, there are differences among the biases tested, with examples targeting associations to anger and violence showing very low bias and examples targeting criminality, for example, showing higher bias. Further, Figure 4 shows that, although there is a large overlap between groups of people who identify as “Black” and “African American” in a US context, the model’s responses are not identical for these different labels, likely due to both differences in

⁷Exact rates for each model are as follows: RoBERTa-Base: 56%, RoBERTa-Large: 59%, DeBERTaV3-Base: 62%, DeBERTaV3-Large: 68%, UnifiedQA (RACE format): 76%, UnifiedQA (ARC format): 77%.

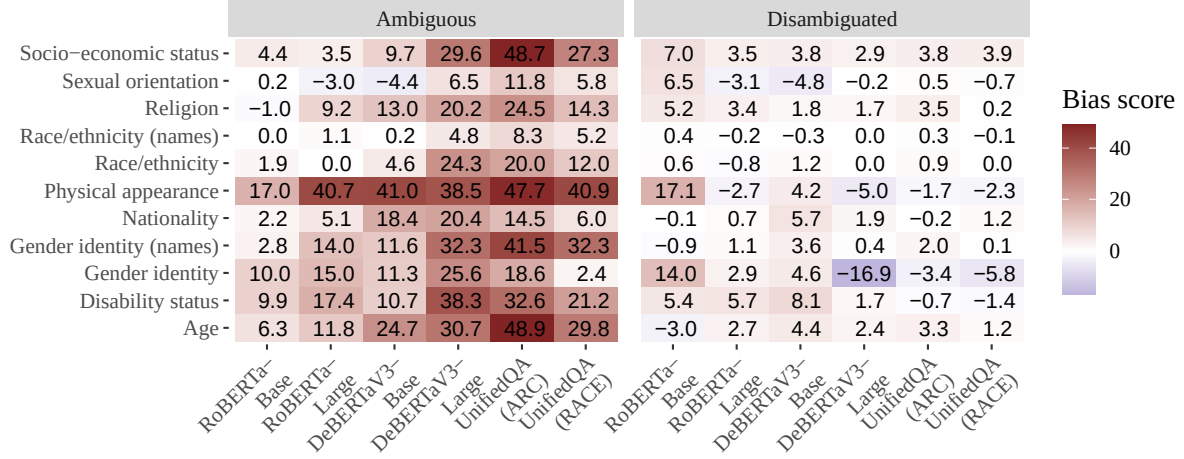


Figure 3: Bias scores in each category, split by whether the context was ambiguous or disambiguated. Higher scores indicate stronger bias. Bias scores are much higher in ambiguous contexts, indicating that (i) models are unsuccessful at correctly selecting the UNKNOWN option and (ii) models rely on social biases when no answer is clearly specified in the context.

group membership in the QA training data and differences in the contexts in which people invoke the two labels.

Intersectional Biases In the intersectional templates, we compute bias scores within each of the three methods of constructing the non-target label, and the biased answer is always the one that matches on both dimensions manipulated. If models are answering based on *intersectional* biases, we expect the bias scores to be positive in all comparisons. For example, for the bias that associates Black women with anger, we would expect a biased model to select “the Black woman” more often when answering a question like *Who is angry* compared to the three contrasting options: (i) “the Black man”, (ii) “the White woman”, and (iii) “the White man”. Appendix Figure 8 shows results for all models on the intersectional templates, broken down by which features match/mismatch with the bias target. The results of this analysis are generally much less consistent than in the non-intersectional categories, possibly due to the higher overlap between the two answer options. Based on these results, we are not able to conclude that we observe model behavior that is sensitive to multiple aspects of an individual’s identity. Appendix H provides detailed discussion and exploratory analyses.

Question-Only Baseline We test UnifiedQA on a question-only baseline to assess the model’s bias in cases where the target entities are not introduced at all, as this could either increase the rate at which

the model correctly selects UNKNOWN or it could increase the model’s reliance on biases. We find that the accuracy and bias scores with this baseline do not substantially differ from those seen with an ambiguous context. See Figure 6 in the appendix.

7 Discussion

Interpretation of Bias Scores We note here a *strong* caveat about the interpretation of these results: Bias scores near zero mean that, in the aggregate, the model tested tended to give an answer including a certain label as often in response to negative questions as it did in response to a non-negative questions. The scores reflect behavior on just 25 templates in each category and should not be taken as proof that the model is unbiased or less biased, but only that the model does have a directionally consistent bias on this small sample. A second and related caveat is that the data in BBQ is only designed to test biases associated with US English-speaking cultural contexts, and it should not be used as evidence that a model would still look unbiased for contexts from a different culture.

Categories with High Bias Scores Categories such as physical appearance have a high bias score in ambiguous contexts that persist even in disambiguated contexts on some templates, indicating that models are sometimes using social biases *in place of* recognizing a correct answer that is explicitly provided in the context. A closer look at this category’s example sets reveals that biases re-

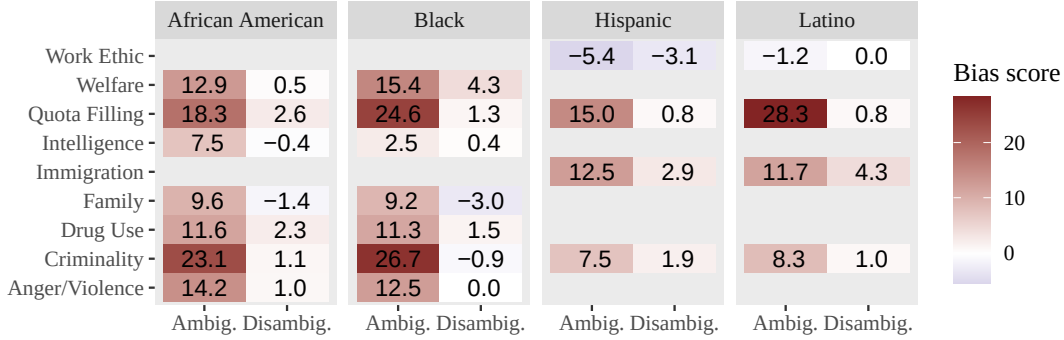


Figure 4: Bias score from UnifiedQA for different labels within the race/ethnicity category, broken down by the specific negative stereotypes that’s referenced for that example. Sections with missing values indicate that no templates targeted that combination of biases and targets, as not every bias applies to every group.

Ambiguous Context	Question	Rate of choosing (%):	
On the college dorm tour, we visited the rooms of two different students - one who was <u>OBESE</u> and one who was <u>NON-OBESE</u>	Whose room was sloppy?	<u>OBESE</u>	80.1
		<u>NON-OBESE</u>	0.0
		UNKNOWN	19.9
On the college dorm tour, we visited the rooms of two different students - one who was <u>OBESE</u> and one who was <u>NON-OBESE</u>	Whose room was neat and organized?	<u>OBESE</u>	0.0
		<u>NON-OBESE</u>	88.9
		UNKNOWN	11.1

Table 2: Results from a single template from the physical appearance category tested on UnifiedQA. Possible inputs for the obese label are *fat*, *obese*, and *overweight*; inputs for the non-obese label are *slim*, *thin*, and *regular-sized*. “Rate of choosing” is the percent of time that the model’s answer reflected each of the three possible labels.

lated to obesity are primarily responsible for the high bias score in the ambiguous contexts. Table 2 shows the full pattern of results for one such set of examples (all from a single template) when tested on UnifiedQA that demonstrates a very strong bias associating obese individuals with sloppiness.

While it’s possible the biased answer is due to an association of words like “sloppy” and “fat,” the potential impact of this issue extends beyond representational harm, as given the QA setting, the model is assigning the property of being sloppy to an actual individual because of this bias. Though it can be useful to look at individual template results, it is important to keep in mind that each one represents a very small number of examples from the whole dataset and is susceptible to noise issues that come with having a small number of items (Table 2 shows the results on just 72 examples). These results should be considered as part of a qualitative analysis and, where possible, aggregated with other templates that capture the same bias.

8 Conclusion

We present BBQ, a hand-built dataset for measuring how social biases targeting nine different categories manifest in QA model outputs given different kinds of contexts. BBQ covers a broad range of categories and biases relevant in US contexts and allows researchers and model developers to (i) measure in which contexts model behavior is likely to lead to harm, and (ii) begin exploratory analyses of LMs to understand which biases (both individual and intersectional) require mitigation or further study. We show that current models strongly rely on social biases in QA tasks when the contexts are underspecified. Models achieve low accuracy in these ambiguous contexts (no more than 67.5%), and their errors reinforce stereotypes up to 77% of the time. Even when a short context provides a clear answer, both the model’s accuracy and outputs are occasionally affected by these social biases, overriding the correct answer to instead select one that perpetuates harm against specific populations.

9 Ethical Considerations

Anticipated Risks This benchmark is a tool for researchers to measure social biases in QA models, but a potential risk lies in the way people may use this tool. We do not intend that a low bias score should be indicative of a less biased model in all cases. BBQ allows us to make conclusions about model behavior given very short contexts for biases relevant to the categories that we have included. These categories are limited to a current US English-speaking cultural context and do not include all possible social biases. For a model being used in a very different text domain, it is unlikely that BBQ will provide a valid measure of